PHYS 230 - Electricity and Magnetism
Lecture 23

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Outline

5. Current, Resistance, and Electromotive Force
   - Electromotive Force (Continued)
   - Circuit Elements and Loops
   - Energy and Power in Circuits

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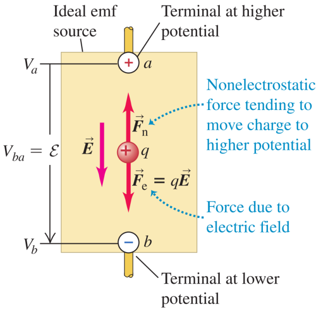
Source of emf

Inside the battery, and circuit not closed: **E** is from + to −

Figure 25.13 Schematic diagram of a source of emf in an "open-circuit" situation. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q.



When the emf source is not part of a closed circuit, $F_n = F_e$ and there is no net motion of charge between the terminals.

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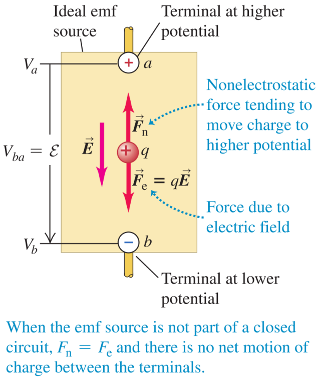

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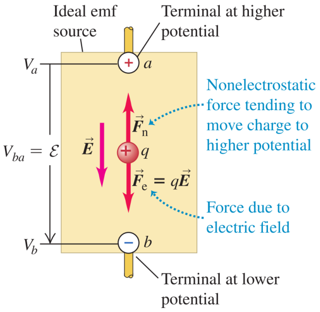

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But there is another nonelectrostatic force $\mathbf{F}_n$ acting on charges moving them "uphill" from $-$ to $+$

When $\mathbf{F}_n$ moves q from $-$ to $+$, the electric potential energy of q increases since

$$U_+ - U_- = q\left(V_+ - V_-\right) \Rightarrow U_+ = qV_{-+} + U_-$$

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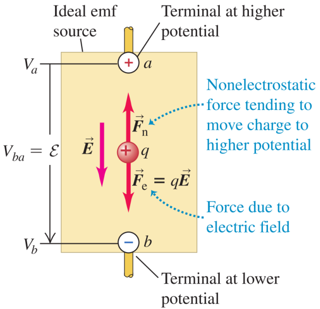

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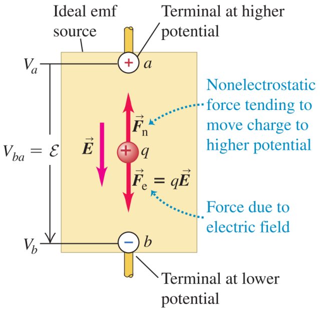

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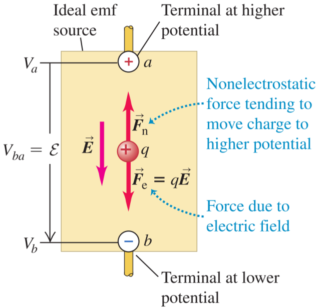

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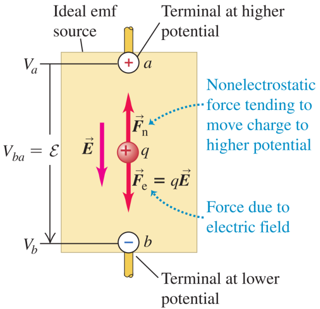

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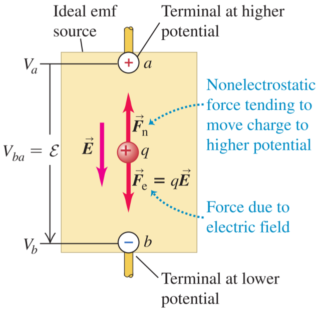

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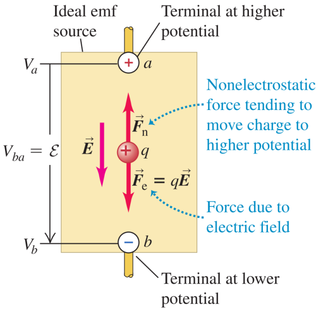

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Dividing by charge q we get

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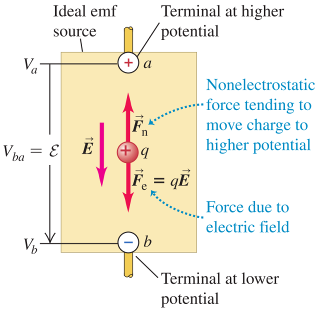

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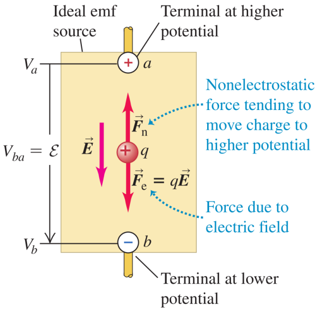

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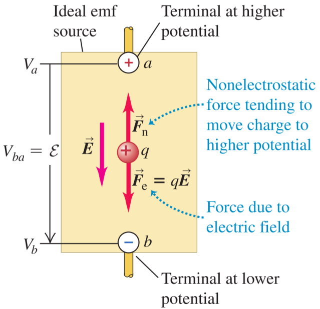

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$$\mathcal{E} = V_{-+} \qquad (\text{or } \mathcal{E}{=}V_{ba})$$

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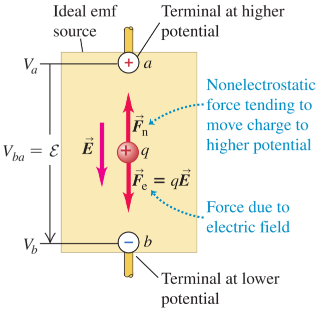

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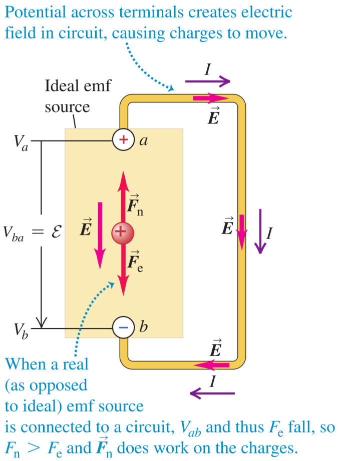

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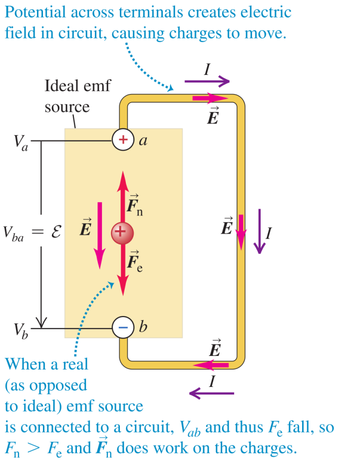

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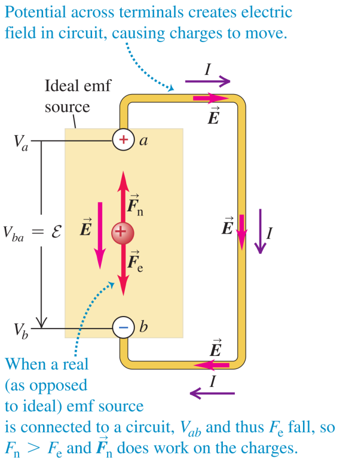

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Outside the battery in the wire we have

$$\underbrace{V}_{V_+ - V_-} = IR$$

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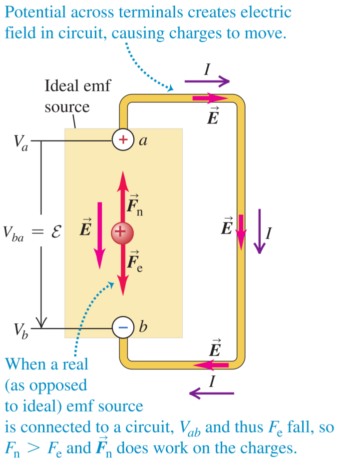

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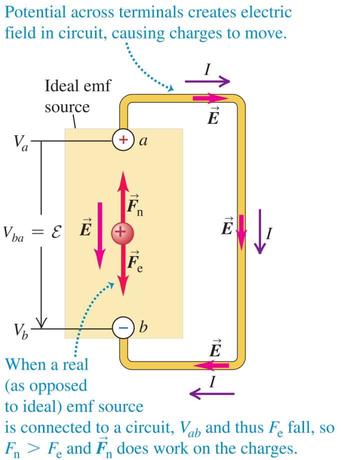

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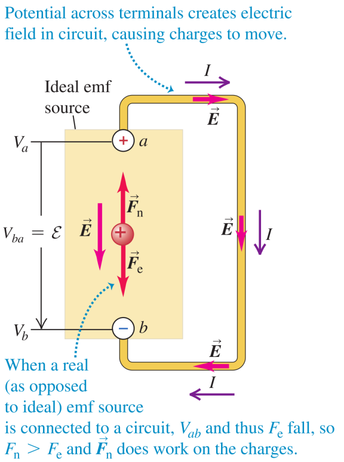

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When a positive charge q flows around the circuit, the potential rises by $\mathcal{E}$ as q passes through the ideal battery and decreases by $V_{-+} = RI$ as q moves through the rest of the circuit; thus numerically $\mathcal{E} = IR$

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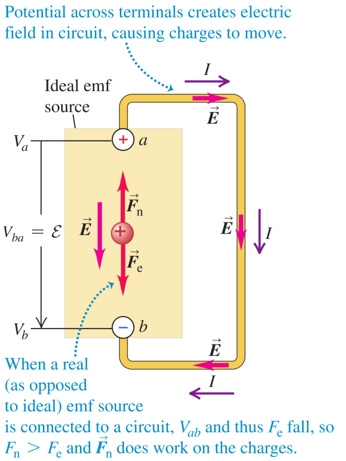

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Real battery: q goes from $-$ to $+$, encounters <u>internal</u> resistance r

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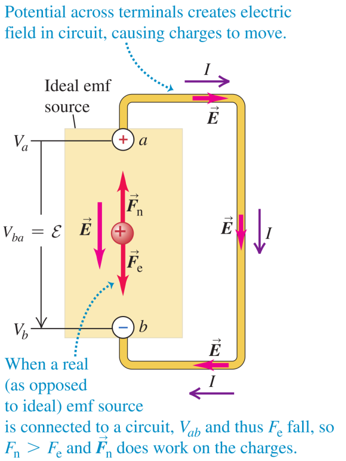

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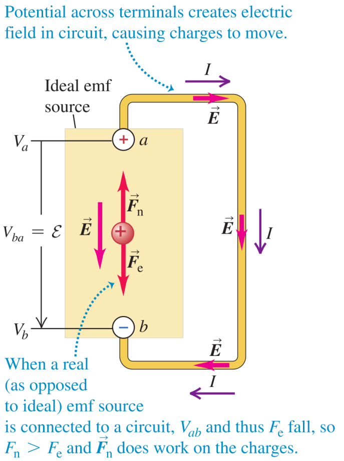

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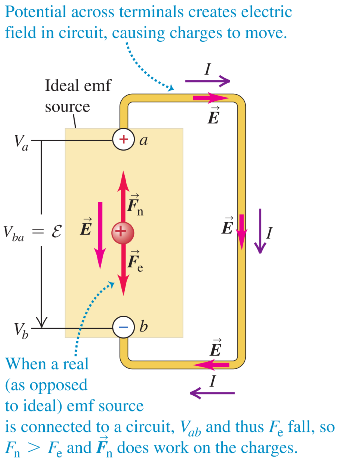

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The energy per unit charge (voltage) needed to overcome r is the current in the battery (and the circuit) I, times r; so the net energy per unit charge (voltage) added to q going from $-$ to $+$ in the battery is

$$\boxed{V_{-+} = \mathcal{E} - Ir}$$

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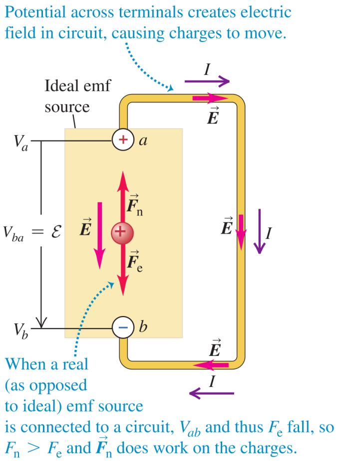

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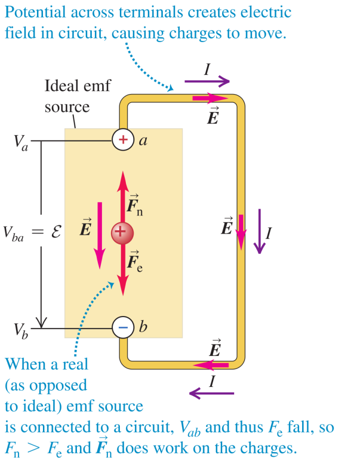

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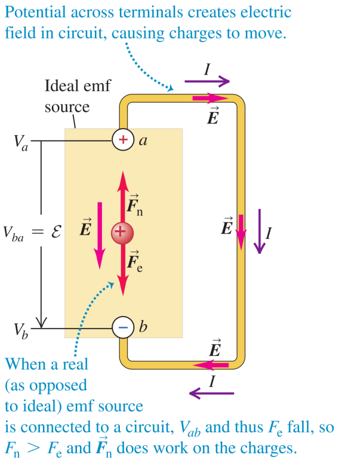

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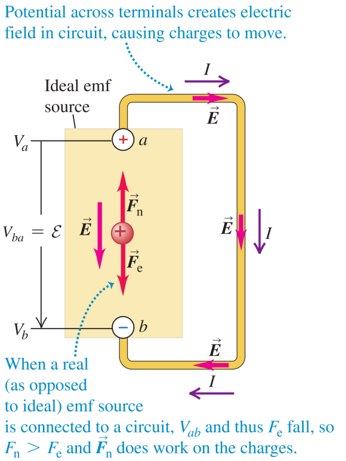
And this will be the potential difference between a and b in the rest of the circuit so

$$\begin{cases} V_{-+} = \mathcal{E} - Ir, & \text{From circuit part outside battery} \\ V_{-+} = IR, & \text{From circuit part inside battery} \end{cases}$$

Figure 25.14 Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q. The current is in the direction from a to b in the external circuit and from b to a within the source.



Real Source of emf

Real battery: q goes from $-$ to $+$, encounters internal resistance r

Hence moving from $-$ to $+$, some of the energy per unit charge (i.e. voltage) of the battery is used to overcome this resistance;

The energy per unit charge (voltage) needed to overcome r is the current in the battery (and the circuit) I, times r; so the net energy per unit charge (voltage) added to q going from $-$ to $+$ in the battery is

$$\boxed{V_{-+} = \mathcal{E} - Ir}$$

And this will be the potential difference between a and b in the rest of the circuit so

$$\begin{cases} V_{-+} = \mathcal{E} - Ir, & \text{From circuit part outside battery} \\ V_{-+} = IR, & \text{From circuit part inside battery} \end{cases} \Rightarrow \boxed{\mathcal{E} - Ir = IR}$$

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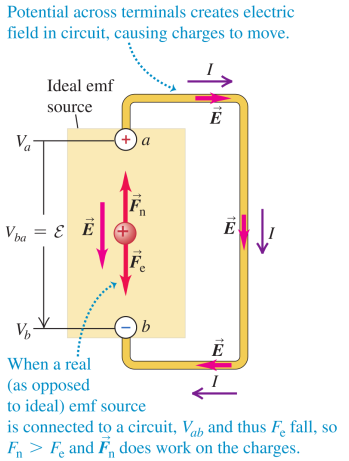

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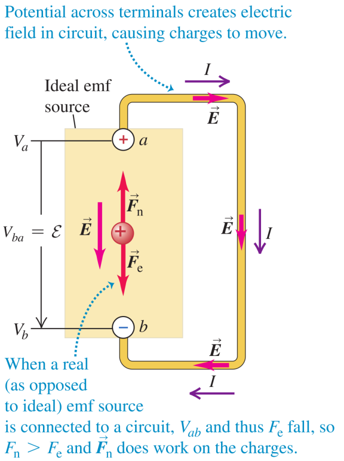
V_{-+} is called the **terminal voltage** and $\mathcal{E}$ is the emf

Figure **25.14** Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q. The current is in the direction from a to b in the external circuit and from b to a within the source.



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In real batteries, when they are advertised as , e.g., 1.5 V, they are referring to the V_{-+} when there is no current

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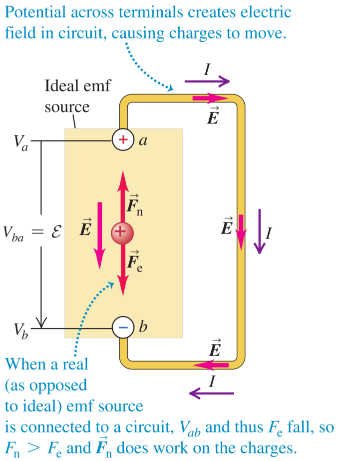

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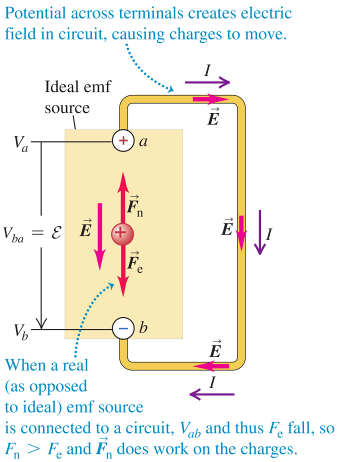

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$$V_{-+} = \mathcal{E} - Ir \overset{I=0}{\Rightarrow} V_{-+} = \mathcal{E}$$

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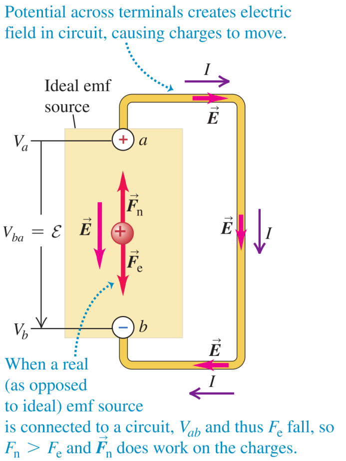

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so they are referring to $\mathcal{E} = 1.5$ V

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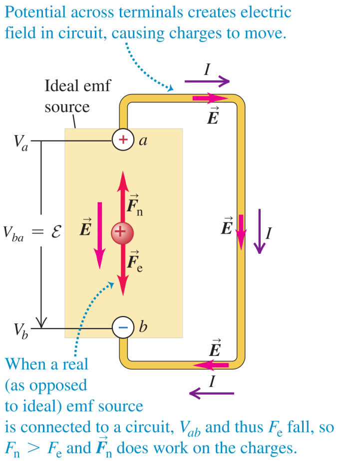

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After the circuit is closed and the current is flowing the terminal voltage is less

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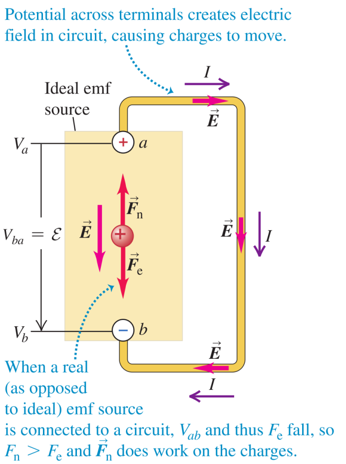

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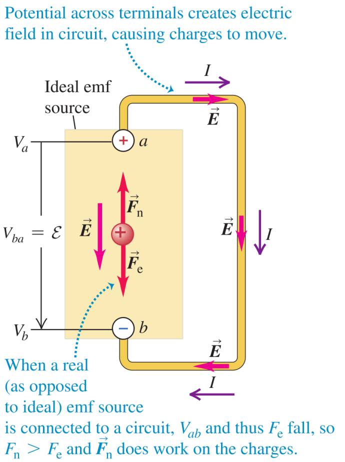

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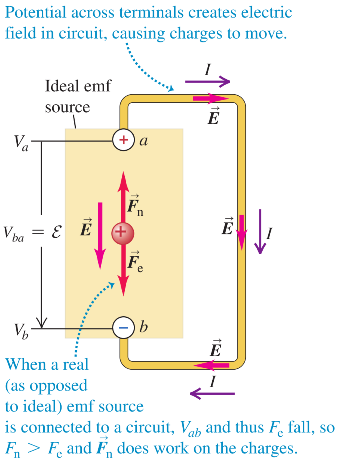
so they are referring to $\mathcal{E} = 1.5$ V

After the circuit is closed and the current is flowing the terminal voltage is less

$$V_{-+} = \mathcal{E} - Ir \overset{I\neq 0}{\Rightarrow} V_{-+} = \mathcal{E} - Ir < \mathcal{E}$$

Figure **25.14** Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q. The current is in the direction from a to b in the external circuit and from b to a within the source.



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In real batteries, when they are advertised as , e.g., 1.5 V, they are referring to the V_{-+} when there is no current

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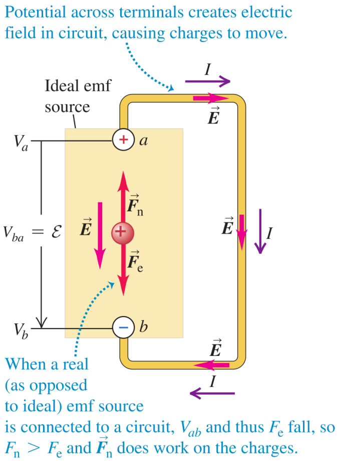
After the circuit is closed and the current is flowing the terminal voltage is less

$$V_{-+} = \mathcal{E} - Ir \overset{I\neq 0}{\Rightarrow} V_{-+} = \mathcal{E} - Ir < \mathcal{E}$$

The internal resistance r of a commercial 12 V lead storage battery is only a few thousandths of an ohm.

Figure **25.14** Schematic diagram of an ideal source of emf in a complete circuit. The electric-field force $\vec{F}_e = q\vec{E}$ and the nonelectrostatic force $\vec{F}_n$ are shown for a positive charge q. The current is in the direction from a to b in the external circuit and from b to a within the source.



Outline

5. Current, Resistance, and Electromotive Force
   - Electromotive Force (Continued)
   - Circuit Elements and Loops
   - Energy and Power in Circuits

All figures in these slides are from one of the following sources, unless cited otherwise:

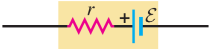
University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Symbols For Circuit Diagrams

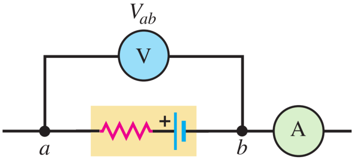
TABLE 25.4 Symbols for Circuit Diagrams

	Conductor with negligible resistance
	Resistor
	Source of emf (longer vertical line always represents the positive terminal, usually the terminal with higher potential)
 or 	Source of emf with internal resistance r (r can be placed on either side)
	Voltmeter (measures potential difference between its terminals)
	Ammeter (measures current through it)

CAUTION Internal resistance is an intrinsic part of a source of emf In Table 25.4 we've drawn the emf $\mathcal{E}$ and internal resistance r as distinct parts of a source of emf. In fact, both $\mathcal{E}$ and r are always present in any source. They *cannot* be separated.

Ammeter and Voltmeter

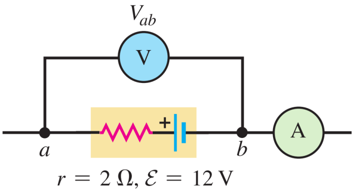
- **Ammeter:**



$r = 2\ \Omega,\ \mathcal{E} = 12\ \text{V}$

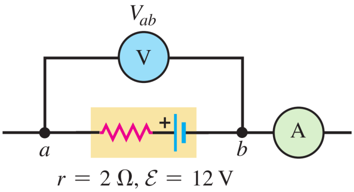
Ammeter and Voltmeter

- **Ammeter:**
  - measures currents



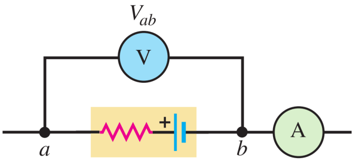
Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
  - connects in series



Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
  - connects in series
  - Ideally has zero resistances; no potential difference between its terminals



$r = 2\ \Omega$, $\mathcal{E} = 12\ \mathrm{V}$

Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
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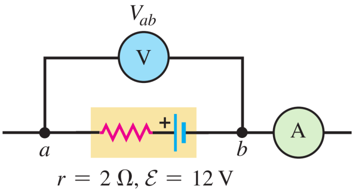
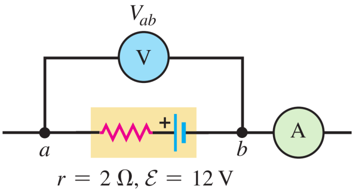
- **Voltmeter:**



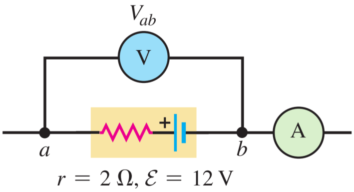
Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
  - connects in series
  - Ideally has zero resistances; no potential difference between its terminals
- **Voltmeter:**
  - measures potential difference between two points in the circuit



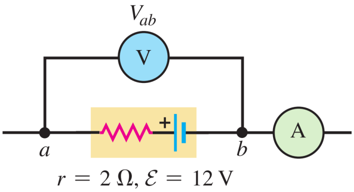
Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
  - connects in series
  - Ideally has zero resistances; no potential difference between its terminals
- **Voltmeter:**
  - measures potential difference between two points in the circuit
  - connects in parallel with some device



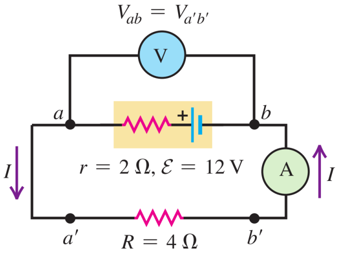
Ammeter and Voltmeter

- **Ammeter:**
  - measures currents
  - connects in series
  - Ideally has zero resistances; no potential difference between its terminals
- **Voltmeter:**
  - measures potential difference between two points in the circuit
  - connects in parallel with some device
  - Ideally has infinite resistance resistances; no currents passes through it



Exercise 1 https://epoll.srv.ualberta.ca – Code: MBD

In figure below, determine the respective readings V_{ba} and I of the idealized voltmeter V and the idealized ammeter A.



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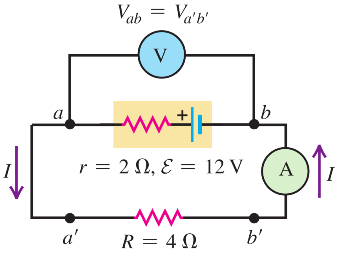
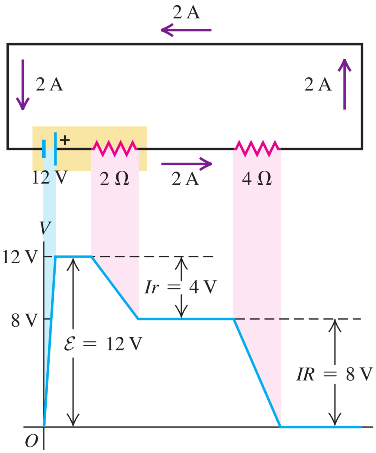


Analyzing Each Loop in a Circuit

1. Choose a certain loop in the circuit (here there is only one loop)

Figure **25.20** Potential rises and drops in a circuit.



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2. Current I in the circuit outside the battery always goes from $+$ terminal of battery to $-$ terminal

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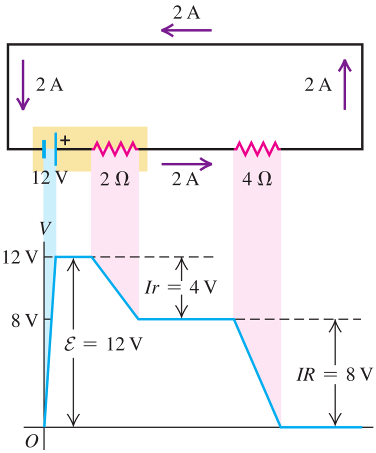

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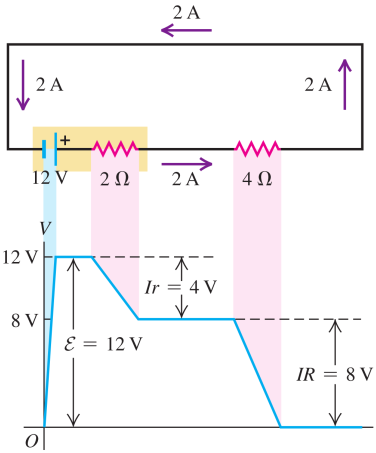

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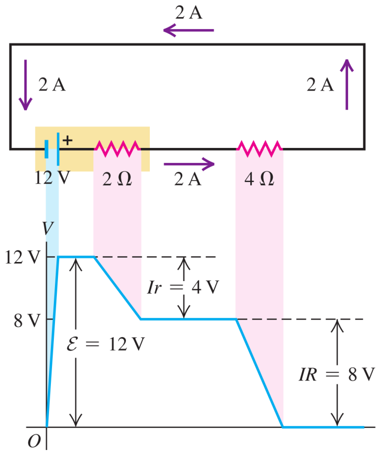

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5. In going around the loop (since you are going from higher V_+ to lower V_-, except in the battery):

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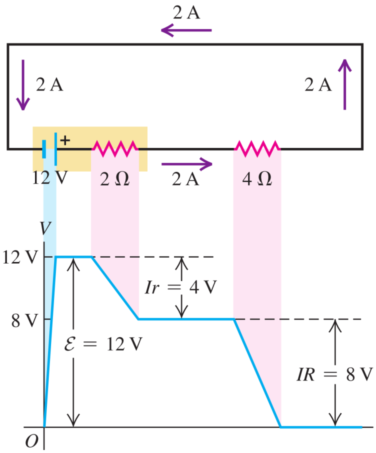

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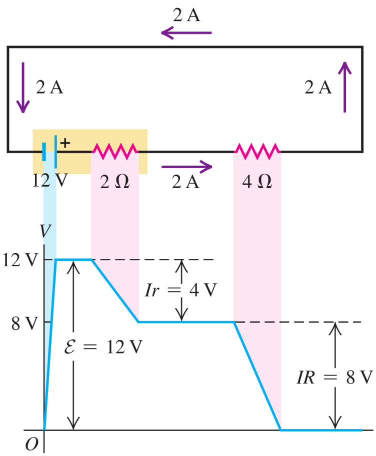

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   5.1 When passing through any non-battery device, V decreases by the potential of that device, e.g., IR
   5.2 When passing through battery, V increases (you are going against I) by $\mathcal{E} - Ir$ (you are going from lower V_- to higher V_+)

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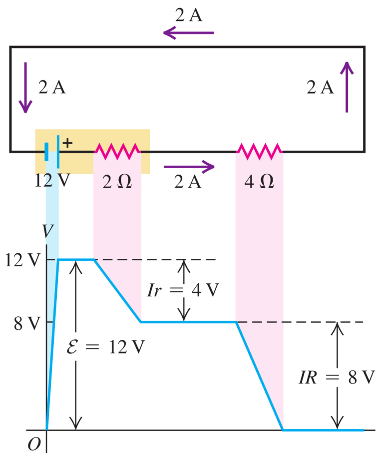
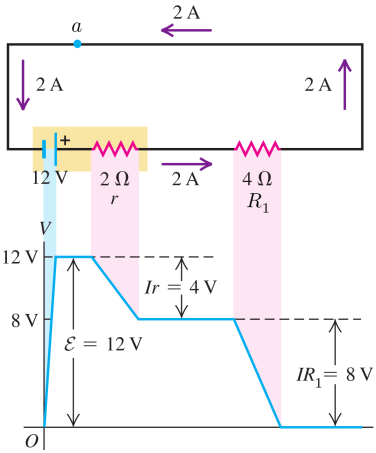


Analyzing Each Loop in a Circuit

We have

V_a

Figure **25.20** Potential rises and drops in a circuit.

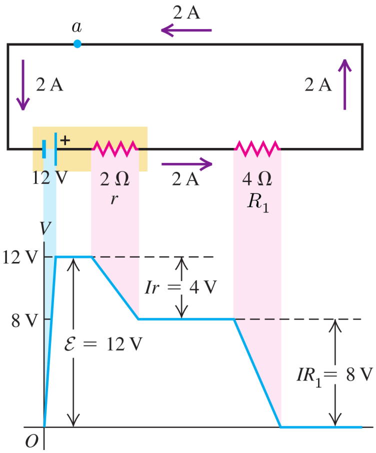


Analyzing Each Loop in a Circuit

We have

$$V_a + (\mathcal{E} - Ir)$$

Figure **25.20** Potential rises and drops in a circuit.



Analyzing Each Loop in a Circuit

We have

$$V_a + (\mathcal{E} - Ir) - IR_1$$

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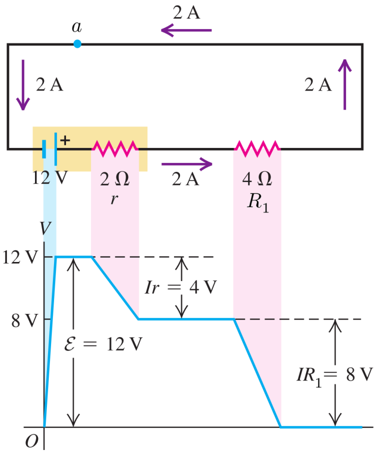

Analyzing Each Loop in a Circuit

We have

$$V_a + (\mathcal{E} - Ir) - IR_1 = V_a$$

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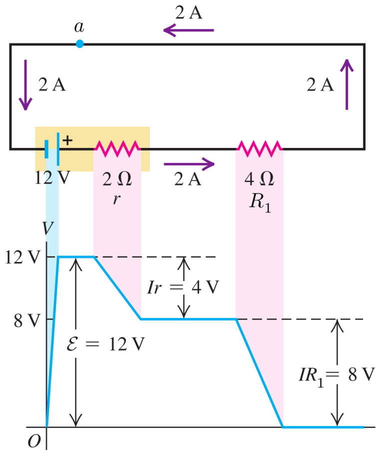


Analyzing Each Loop in a Circuit

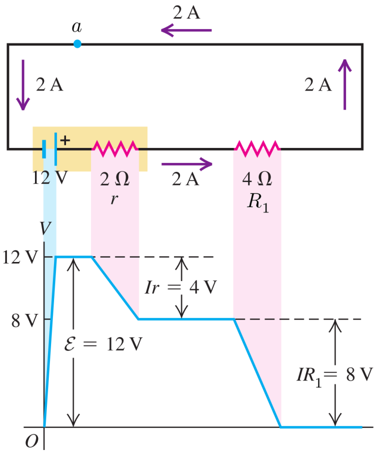
We have

$$V_a + (\mathcal{E} - Ir) - IR_1 = V_a$$

And hence

$$\mathcal{E} - Ir - IR_1 = 0$$

Figure 25.20 Potential rises and drops in a circuit.



Analyzing Each Loop in a Circuit

We have

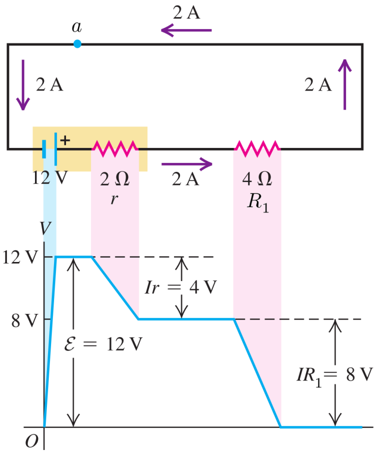
$$V_a + (\mathcal{E} - Ir) - IR_1 = V_a$$

And hence

$$\mathcal{E} - Ir - IR_1 = 0$$

We can solve this equation for any of $\mathcal{E}$, I, r, R_1

Figure **25.20** Potential rises and drops in a circuit.



Outline

5. Current, Resistance, and Electromotive Force
    - Electromotive Force (Continued)
    - Circuit Elements and Loops
    - Energy and Power in Circuits

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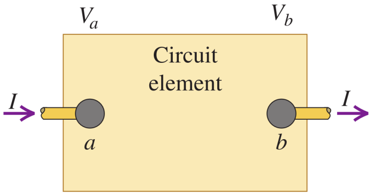
Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

Power in a Circuit

Consider a circuit element as in figure with potential difference ($V_a > V_b$)

$$V = V_a - V_b > 0$$



Power in a Circuit

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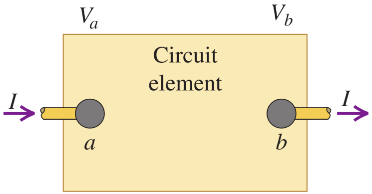
A charge dq going from a to b (in the direction of I) experiences a decrease in potential energy

$$dU = dq\left(V_b - V_a\right) \Rightarrow dU = -Vdq$$



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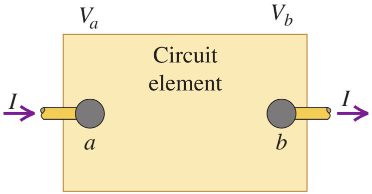

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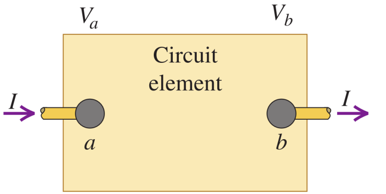

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dq going from a to b releases (negative sign below!) energy per unit time, with power

$$P = \frac{dU}{dt} = -V \underbrace{\frac{dq}{dt}}_{I} \Rightarrow P = -VI$$

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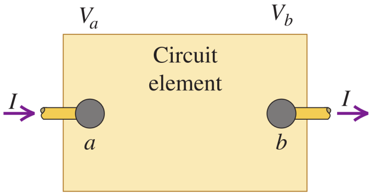

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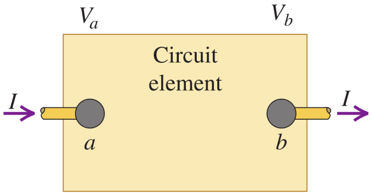

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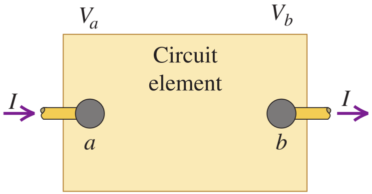

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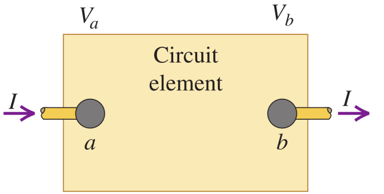

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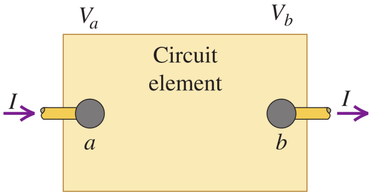
Power in a Circuit

In general

$$\boxed{P = VI}$$



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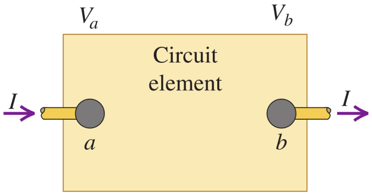

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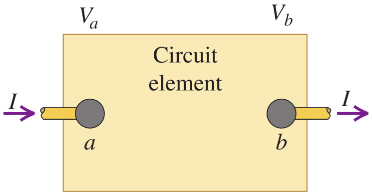

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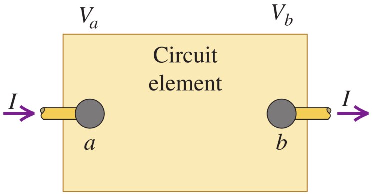

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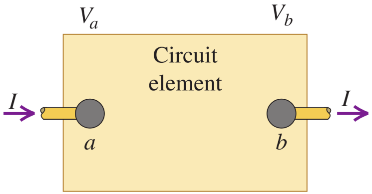
Power in a Circuit of an element with R

Consider a circuit element as in figure with potential difference $V = V_a - V_b > 0$ and resistance R



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Consider a circuit element as in figure with potential difference $V = V_a - V_b > 0$ and resistance R

Since we have $V = RI$, the power due to this object is

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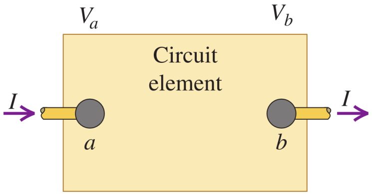

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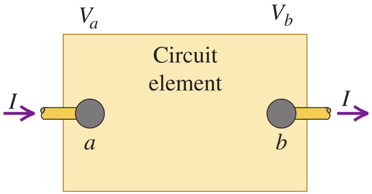

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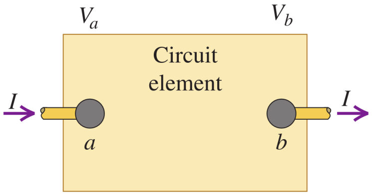

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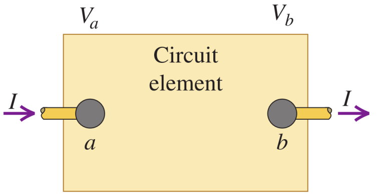


Except in battery, charges lose potential energy (negative power)

But they obtain kinetic energy K due to conservation of energy

Then they hit other charges and stationary atoms, vibrate them $\Rightarrow$ energy (K) released out to environment as heat

Power Output of a Source of emf

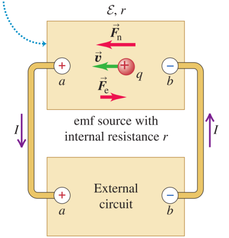
In the battery shown, a positive charge goes from $-$ (terminal b) to $+$ (terminal a), gains energy (since $V_{ba} = V_a - V_b > 0$) with the power

$$P = V_{ba} I$$

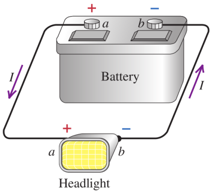
Figure 25.22 Energy conversion in a simple circuit.

(a) Diagrammatic circuit

- The emf source converts nonelectrical to electrical energy at a rate $\mathcal{E}I$.
- Its internal resistance *dissipates* energy at a rate $I^2 r$.
- The difference $\mathcal{E}I - I^2 r$ is its power output.



(b) A real circuit of the type shown in (a)



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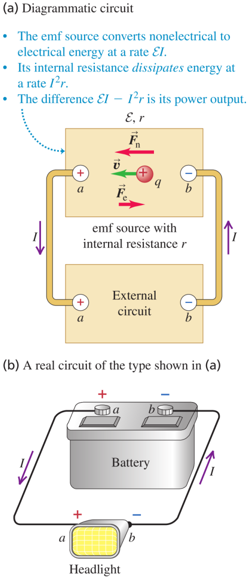

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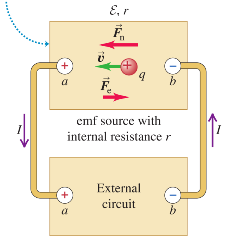

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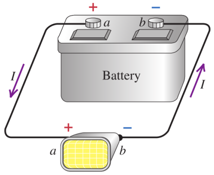

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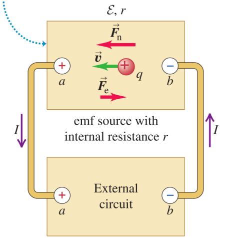

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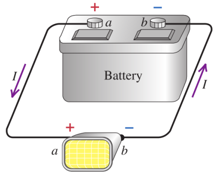

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$I^2 r$ is the rate of dissipation(loss)/transfer of its potential energy due to collision to other atoms

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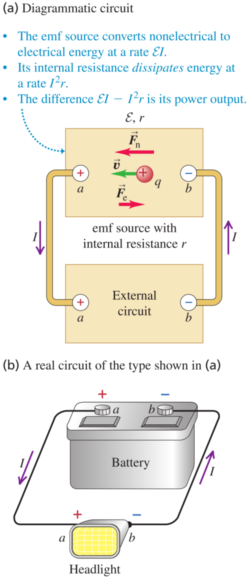

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$I^2 r$ is the rate of dissipation(loss)/transfer of its potential energy due to collision to other atoms

At b it has gained potential energy at a rate $P = \mathcal{E} I - I^2 r$ which is then delivered/released to the circuit as kinetic energy

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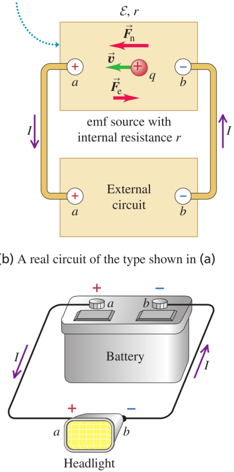

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Power Output of a Source of emf

The battery's net power output is

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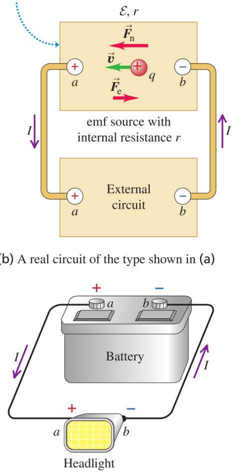

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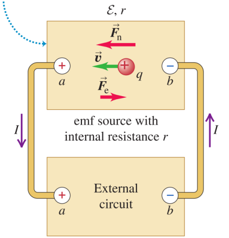

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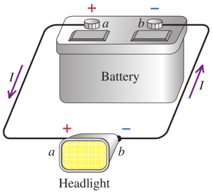

Power Output of a Source of emf

The battery's net power output is

$$P = \mathcal{E}I - I^2 r$$

Its power dissipation is

$$I^2 r$$

and its rate (or power) of energy conversion is

$$\mathcal{E}I$$

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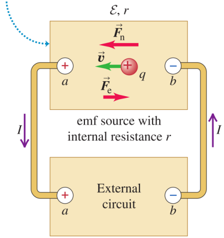

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